How-To Library

Informatica

Address Verification Best Practices for Hong Kong

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Abstract

This document tells you how to connect the elements in a Hong Kong address to the appropriate address elements in Informatica Address Verification. This document also describes best practices that you can implement when you verify addresses in Hong Kong.

Supported Versions

- Informatica Address Verification 3.0 and later
- Informatica Address Verification (On-Premises) 5.x

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Introduction

Informatica Address Verification can validate and enhance addresses in Hong Kong. This how-to article provides you with configuration information and best practices that you can follow to get the most from the addresses. The best practices include advice on how to select the appropriate XML elements for Hong Kong address data.

To verify the quality of an input address, Address Verification compares the address information to the data in one or more reference data files. You can configure Address Verification to analyze the different types of address information in the input data. You must map each item of address information to an appropriate element in Address Verification.

When you select the input address elements, you must ensure that the elements are correct for the different types of input information. The country that the address identifies determines the elements that you need.

Process Modes

Informatica Address Verification supports the following process modes for Hong Kong addresses:

- BATCH
- COUNTRY RECOGNITION
- FAST COMPLETION
- INTERACTIVE

- PARSE

Character Set

Hong Kong reference address databases store address data in the native Chinese and Latin scripts. By default, Informatica Address Verification returns the verified Hong Kong addresses in the native script.

Address Verification reads and writes Chinese-language characters in the Hanzi script.

Address Format and Address Structure

You improve the performance of Address Verification when you define the input address in the format that the postal service expects.

The address structure that you define depends on the character set that you use. A Hong Kong address in the native script presents the address elements in the reverse order to a Hong Kong address in the Latin script.

Address Structure for Latin Characters

The Hong Kong postal service recognizes the following structure for a postal address in the Latin script:

```
Line 1: RECIPIENT OR CONTACT NAME
Line 2: [SUB-BUILDING INFORMATION]
Line 3: [BUILDING INFORMATION]
Line 4: HOUSE NUMBER STREET NAME
Line 5: CITY
Line 6: PROVINCE
Line 7: COUNTRY
```

The following sample address illustrates the address in the Latin script:

```
12/F, 1201 ROOM
CHEONG HING BUILDING
72 NATHAN ROAD
TSIM SHA TSUI
KOWLOON
HKG
```

Address Structure for Chinese Characters

A Hong Kong address in the native script begins with the country element, and the elements continue from the largest geographical element to the smallest geographical element.

The Hong Kong postal service recognizes the following structure for a postal address in the native script:

```
Line 1: COUNTRY
Line 2: PROVINCE
Line 3: CITY
Line 4: STREET NAME HOUSE NUMBER
Line 5: [BUILDING INFORMATION]
Line 6: [SUB-BUILDING INFORMATION]
Line 7: RECIPIENT OR CONTACT NAME
```

The following sample address illustrates the address in the native script:

```
HKG
九龍
尖沙咀
彌敦道 72
昌興大廈
12 樓, 1201 室
```

Key Address Elements

Address Verification is unlikely to verify an address successfully if one of the following items is absent from the address:

- Locality
- Street

Mapping Input Address Data to Address Elements

To verify an address, map the input address information to appropriate elements in Address Verification.

The following table shows how you can map the input address to the address elements:

Information Type	Address Information	Address Elements
Street Name	彌敦道	Street
House Number	72	HouseNumber
Building Name	昌興大廈	Building
Floor	12 樓	SubBuilding1
Apartment	1201 室	SubBuilding2
City	尖沙咀	Locality
Administrative Region	九龍	Province
Country	HKG	Country

Input and Result XML in Address Verification (On-Premises)

The following example shows how you can map the address information to the XML address elements in Address Verification (On-Premises).

You enter the following input address:

```
<InputData>
    <AddressElements>
        <Country Item="1" Type="NAME">HKG</Country>
        <Locality Item="1" Type="COMPLETE">尖沙咀</Locality>
        <Province Item="1" Type="COUNTRY_STANDARD">九龍</Province>
        <Street Item="1" Type="COMPLETE">彌敦道</Street>
        <Number Item="1" Type="COMPLETE">72</Number>
        <Building Item="1" Type="COMPLETE">昌興大廈</Building>
        <SubBuilding Item="1" Type="COMPLETE">12 樓</SubBuilding>
        <SubBuilding Item="2" Type="COMPLETE">1201 室</SubBuilding>
    </AddressElements>
</InputData>
```

Informatica Address Verification (On-Premises) returns the following result:

```
<?xml version="1.0" encoding="UTF-16"?>

<!-- EngineVersion="5.16.2.60959" -->

<Result     ProcessStatus="V4"
     ModeUsed="BATCH"
     Count="1"
```

```
     CountOverflow="NO"
     CountryISO3="HKG"
     PreferredScript="DATABASE"
     PreferredLanguage="DATABASE">

<ResultData ResultNumber="1"
     MailabilityScore="5"
     ResultPercentage="100.00"
     ElementResultStatus="00F0F8F0F000F0FF00E0"
     ElementInputStatus="00606060600060660060"
     AddressResolutionCode="00000000000000000000"
     ExtElementStatus="00505550000505500000"
     ElementRelevance="00101010100010000010"
     AddressType="B"
     LanguageISO3="ZHO">
    <AddressElements>
        <Country Type="NAME_EN" Item="1">HONG KONG</Country>
        <Locality Item="1">尖沙咀</Locality>
        <Province Item="1">九龍</Province>
        <Province Item="2">油尖旺</Province>
        <Street Item="1">彌敦道</Street>
        <Number Item="1">72</Number>
        <Building Item="1">昌興大廈</Building>
        <SubBuilding Item="1">12 樓</SubBuilding>
        <SubBuilding Item="2">1201 室</SubBuilding>
    </AddressElements>
    <AddressLines>
        <DeliveryAddressLine Line="1">彌敦道 72</DeliveryAddressLine>
        <DeliveryAddressLine Line="2">昌興大廈 </DeliveryAddressLine>
        <DeliveryAddressLine Line="3">12 樓, 1201 室</DeliveryAddressLine>
        <CountrySpecificLocalityLine Line="1">尖沙咀</CountrySpecificLocalityLine>
        <FormattedAddressLine Line="1">九龍</FormattedAddressLine>
        <FormattedAddressLine Line="2">尖沙咀</FormattedAddressLine>
        <FormattedAddressLine Line="3">彌敦道 72</FormattedAddressLine>
        <FormattedAddressLine Line="4">昌興大廈 </FormattedAddressLine>
        <FormattedAddressLine Line="5">12 樓, 1201 室</FormattedAddressLine>
    </AddressLines>
    <AddressComplete>九龍
尖沙咀
彌敦道 72
昌興大廈
12 樓, 1201 室</AddressComplete>
</ResultData>
</Result>
```

Input and Result XML in Address Verification

The following example shows how you can verify a Hong Kong address in a SOAP call to Address Verification.

You submit the following request:

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:v4="http://
validator5.AddressDoctor.com/Webservice5/v4">
   <soapenv:Header/>
   <soapenv:Body>
      <v4:Process>
         <v4:login>....</v4:login>
         <v4:password>....</v4:password>
         <v4:parameters>
            <v4:ProcessMode>Batch</v4:ProcessMode>
            <v4:ValidationParameters>
            </v4:ValidationParameters>
         </v4:parameters>
         <v4:addresses>
            <v4:Address>
               <v4:Building>
                  <v4:string>昌興大廈</v4:string>
               </v4:Building>
               <v4:SubBuilding>
                  <v4:string>12 樓</v4:string>
```

```
                  <v4:string>1201 室</v4:string>
               </v4:SubBuilding>
               <v4:Street>
                  <v4:string>彌敦道</v4:string>
               </v4:Street>
               <v4:HouseNumber>
                  <v4:string>72</v4:string>
               </v4:HouseNumber>
               <v4:Locality>
                  <v4:string>尖沙咀</v4:string>
               </v4:Locality>
               <v4:PostalCode>
                  <v4:string></v4:string>
               </v4:PostalCode>
               <v4:Province>
                  <v4:string>九龍</v4:string>
               </v4:Province>
               <v4:Country>
                  <v4:string>HKG</v4:string>
               </v4:Country>
            </v4:Address>
         </v4:addresses>
         <v4:enrichments>
         </v4:enrichments>
      </v4:Process>
   </soapenv:Body>
</soapenv:Envelope>
```

Informatica Address Verification returns the following response:

```
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/" xmlns:xsi="http://
www.w3.org/2001/XMLSchema-instance" xmlns:xsd="http://www.w3.org/2001/XMLSchema">
   <soap:Body>
      <ProcessResponse xmlns="http://validator5.AddressDoctor.com/Webservice5/v4">
         <ProcessResult>
            <StatusCode>100</StatusCode>
            <StatusMessage>OK</StatusMessage>
            <Results>
               <Result>
                  <ProcessStatus>V4</ProcessStatus>
                  <CountryISO3>HKG</CountryISO3>
                  <ResultDataSet>
                     <ResultData>
                        <ResultNumber>1</ResultNumber>
                        <MailabilityScore>5</MailabilityScore>
                        <ResultPercentage>100.00</ResultPercentage>
                        <ElementInputStatus>0060606060060660060</ElementInputStatus>
                        <ElementResultStatus>00F0F8F0F000F0FF00E0</ElementResultStatus>
                        <ElementRelevance>0010101010010000010</ElementRelevance>
                        <ExtElementStatus>0050555000050550000</ExtElementStatus>
                        <AddressResolutionCode>00000000000000000000</AddressResolutionCode>
                        <AddressType>B</AddressType>
                        <LanguageISO3>CHN</LanguageISO3>
                        <Address>
                           <Building>
                              <string>昌興大廈</string>
                           </Building>
                           <SubBuilding>
                              <string>12 樓</string>
                              <string>1201 室</string>
                           </SubBuilding>
                           <Street>
                              <string>彌敦道</string>
                           </Street>
                           <HouseNumber>
                              <string>72</string>
                           </HouseNumber>
                           <Locality>
                              <string>尖沙咀</string>
                           </Locality>
                           <Province>
```

```
                                    <string>九龍</string>
                                    <string>油尖旺</string>
                                 </Province>
                                 <Country>
                                    <string>HONG KONG</string>
                                 </Country>
                                 <DeliveryAddressLines>
                                    <string>彌敦道 72</string>
                                    <string>昌興大廈</string>
                                    <string>12 樓, 1201 室</string>
                                 </DeliveryAddressLines>
                                 <CountrySpecificLocalityLine>
                                    <string>尖沙咀</string>
                                 </CountrySpecificLocalityLine>
                                 <FormattedAddress>
                                    <string>HONG KONG</string>
                                    <string>九龍</string>
                                    <string>尖沙咀</string>
                                    <string>彌敦道 72</string>
                                    <string>昌興大廈</string>
                                    <string>12 樓, 1201 室</string>
                                 </FormattedAddress>
                                 <AddressComplete>HONG KONG;九龍;尖沙咀;彌敦道 72;昌興大廈 ;12 樓, 1201 室</
AddressComplete>
                              </Address>
                           </ResultData>
                        </ResultDataSet>
                     </Result>
                  </Results>
               </ProcessResult>
            </ProcessResponse>
         </soap:Body>
      </soap:Envelope>
```


Address Enrichments

You can configure Informatica Address Verification to add enrichments to verified Hong Kong addresses.

The following table lists the types of enrichment that you can add to a Hong Kong address:

Enrichment Type	Description
CAMEO	You can retrieve CAMEO profiles as enrichments for valid Hong Kong addresses. CAMEO profiles are consumer classification systems that provide socio-economic and demographic indicators about the residents at the address.
Standard Geocoding	You can retrieve standard, interpolated geocoordinates for valid Hong Kong addresses.
Arrival Point Geocoding	You can retrieve arrival point geocoordinates for valid Hong Kong addresses.

Frequently Asked Questions

Can Address Verification return address information in multiple languages for Hong Kong?

You can configure Address Verification to return valid Hong Kong addresses in Chinese or in English. The default language is Chinese.

To set the preferred language, configure the `PreferredLanguage` attribute of the Result element in `Parameters.xml`.

The following table describes the `PreferredLanguage` values that you can set for Hong Kong address data:

Option	Description
DATABASE	Default value. Returns all address information in Chinese.
ENGLISH	Returns all address information in English.
ALTERNATIVE_1	Returns all address information in the database language.
ALTERNATIVE_2	Returns all address information in English.
ALTERNATIVE_3	Returns all address information in the database language.
PRESERVE_INPUT	Returns the address information in the input language. If Address Verification cannot return an element in the input language, it returns the element in the database language.

Does Address Verification support single-line address verification for Hong Kong addresses in Fast Completion mode?

Yes. Submit the address in the Chinese language and in the Hanzi script. Address Verification reads the address in the Hanzi script and returns the address suggestions in Hanzi.

Enter a Hong Kong address in the following format:

```
[Province] [Locality] [Street] [House Number] [Building 1] [Building 2] [Sub-building]
```

When you enter a partial address in fast completion mode, Address Verification returns one or more address suggestions for the address that you enter. When you enter a complete or almost complete address in fast completion mode, Address Verification returns a single suggestion for the address that you enter.

To verify single-line addresses, enter the addresses in the `AddressComplete` element that the `InputData.xml` file specifies.

Reference Address Databases

Informatica updates the Hong Kong reference address databases twice a year. You can download the latest database packages from the Informatica Database Download portal. For news about reference address databases and software updates for Address Verification, subscribe to the Informatica DaaS Newsletter from Informatica.

Informatica Address Verification provides the following reference databases for Hong Kong addresses:

- `HKG5BI.MD`. Batch and interactive data.
- `HKG5CA.MD`. CAMEO data.
- `HKG5FC.MD`. Fast completion data.
- `HKG5GC.MD`. Standard geocoding data.
- `HKG5GCAP.MD`. Arrival point geocoding data.

Note: You download databases for Informatica Address Verification (On-Premises). You do not need to download databases for Informatica Address Verification, as Informatica Address Verification uses a cloud interface.

Additional Documentation Resources

To view the latest documentation for Informatica Address Verification, click the following links:

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor

https://network.informatica.com/community/informatica-network/products/data_quality/data-as-a-service/address-doctor-cloud

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